

YUNYONG KO

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RESEARCH INTERESTS	My research interests mainly lie in data mining and machine learning on a various types of data (e.g., graph, hypergraph, text, image, and time-series), with a special focus on exploring knowledge for real-world applications.	
EDUCATION	Hanyang University , Seoul, Korea • <i>Ph.D. in Computer Science</i> – Thesis: Effective Approaches to Distributed Deep Learning: Methods, Analyses, and Evaluation – Advisor: Prof. Sang-Wook Kim Hanyang University , Seoul, Korea • <i>B.S. in Computer Science</i>	Sep. 2013 – Aug. 2021 Mar. 2009 – Aug. 2013
POSITIONS	Chung-Ang University , Seoul, Korea • <i>Assistant Professor, School of Computer Science and Engineering</i>	Mar. 2024 – Present
RESEARCH EXPERIENCES	University of Illionois at Urbana-Champaign , Urbana, IL, USA • <i>Postdoctoral Researcher, Department of Computer Science</i> – Topic: Large-Scale Hypergraph Learning for Real-World Applications – Advisor: Prof. Hanghang Tong Hanyang University , Seoul, Korea • <i>Postdoctoral Researcher, Department of Computer Science</i> – Topic: Optimization Technique for Large-Batch DNN Training – Advisor: Prof. Sang-Wook Kim The Pennsylvania State University , University Park, PA, USA • <i>Visiting Scholar, College of Information Sciences and Technology</i> – Topic: Data Parallelism Approach for Distributed Deep Learning – Advisor: Prof. Dongwon Lee	May. 2022 – Feb. 2024 Sep. 2021 – April. 2022 Oct. 2019 – Feb. 2020
AWARDS & HONORS	Received the Scholarship and Teaching for Engineering Postdocs (STEP) • Grainger College of Engineering (GCOE), University of Illinois at Urbana-Champaign Selected as One of the Best-Ranked Papers • IEEE International Conference on Data Mining (IEEE ICDM) Selected as One of the Spotlight Presentations • ACM International Conference on Information and Knowledge Management (ACM CIKM) Received the Outstanding Ph.D. Dissertation Award • Research Institute of Industrial Science, Hanyang University Received the ACM SIGAPP Student Travel Award • ACM Symposium on Applied Computing (ACM SAC) Awarded the NAVER Ph.D. Fellowship • Naver Corporation Domestic Conference/Journal Awards • Best Paper Awards : KIPS Spring Confernece (2021, 2023), Journal of KIISE (2017) • Best Presentation Award : KIISE KCC (2017)	2023 2021 2021 2021 2019 2017

Preprinted Papers

- [2] CROWN: A Novel Approach to Comprehending Users' Preferences for Accurate Personalized News Recommendation

Yunyong Ko, Seongeun Ryu, and Sang-Wook Kim

arXiv:2310.09401

Full Paper (*Under Review at One of the Top-Tier CS Conferences*)

- [1] Enhancing Hyperedge Prediction with Context-Aware Self-Supervised Learning

Yunyong Ko, Hanghang Tong, and Sang-Wook Kim

arXiv:2309.05798

Full Paper (*Under Review at the IEEE Transactions on Knowledge and Data Engineering*)

International Conference and Journal Papers (* indicates equal contributions)

- [13] HearHere: Mitigating Echo Chambers in News Consumption through an AI-based Web System
Youngseung Jeon, Jaehoon Kim, Sohyun Park, Yunyong Ko, Seongeun Ryu, Sang-Wook Kim, and Kyungsik Han

CSCW 2024 (*The ACM Conference on Computer-Supported Cooperative Work and Social Computing*)

Full Paper

- [12] SAGE: A Storage-Based Approach for Scalable and Efficient Sparse Generalized Matrix-Matrix Multiplication
{Myung-Hwan Jang*, Yunyong Ko*}, Hyuck-Moo Gwon, Ik-Hyeon Jo, Yongjun Park, and Sang-Wook Kim

CIKM 2023 (*The ACM International Conference on Information and Knowledge Management*)

Full Paper (Acceptance Rate $\approx 24\%$)

- [11] KHAN: Knowledge-Aware Hierarchical Attention Networks for Accurate Political Stance Prediction

Yunyong Ko, Seongeun Ryu, Soeun Han, Youngseung Jeon, Jaehoon Kim, Sohyun Park, Kyungsik Han, Hanghang Tong and Sang-Wook Kim

WWW 2023 (*The ACM Web Conference*)

Full Paper (Acceptance Rate $\approx 19.2\%$)

- [10] RealGraph^{GPU}: A High-Performance GPU-Based Graph Engine Toward Large-Scale Real-World Network Analysis

Myung-Hwan Jang, Yunyong Ko, Dongkyu Jeong, Jeong-Min Park, and Sang-Wook Kim

CIKM 2022 (*The ACM International Conference on Information and Knowledge Management*)

Short Paper (Acceptance Rate $\approx 28.3\%$)

- [9] Not All Layers Are Equal: A Layer-Wise Adaptive Approach Toward Large-Scale DNN Training

Yunyong Ko, Dongwon Lee, and Sang-Wook Kim

WWW 2022 (*The ACM Web Conference*)

Full Paper (Acceptance Rate $\approx 17.7\%$)

- [8] D-FEND: A Diffusion-Based Fake News Detection Framework for News Articles Related to COVID-19

Soeun Han, Yunyong Ko, Yusim Kim, Heejin Park, Seongsu Oh, and Sang-Wook Kim

SAC 2022 (*The ACM Symposium on Applied Computing*)

Full Paper (Acceptance Rate $\approx 24\%$)

- [7] MASCOT: A Quantization Framework for Efficient Matrix Factorization in Recommender Systems

{Yunyong Ko*, Jae-Seo Yu*}, Hong-Kyun Bae, Yongjun Park, Dongwon Lee, and Sang-Wook Kim

ICDM 2021 (*The IEEE International Conference on Data Mining*)

Full Paper (Acceptance Rate $\approx 9.9\%$)

Selected as One of the Best-ranked Papers of ICDM 2021 for Fast-track Journal Invitation

- [6] ALADDIN: Asymmetric Centralized Training for Distributed Deep Learning

Yunyong Ko, Kibong Choi, Hyunseung Je, Dongwon Lee, and Sang-Wook Kim

CIKM 2021 (*The ACM International Conference on Information and Knowledge Management*)

Full Paper (Acceptance Rate $\approx 21.7\%$)

Selected as One of the Spotlight Presentations of CIKM 2021

- [5] An In-Depth Analysis of Distributed Training of Deep Neural Networks
Yunyong Ko, Kibong Choi, Jiwon Seo, and Sang-Wook Kim
IPDPS 2021 (*The IEEE International Parallel and Distributed Processing Symposium*)
Full Paper (Acceptance Rate $\approx 24.5\%$)
- [4] Influence Maximization for Effective Advertisement in Social Networks: Problem, Solution, and Evaluation
Suk-Jin Hong, Yunyong Ko, Moon-Jeung Joe, and Sang-Wook Kim
SAC 2019 (*The ACM Symposium on Applied Computing*)
Full Paper (Acceptance Rate $\approx 24.2\%$)
- [3] Efficient and Effective Influence Maximization in Social Networks: A Hybrid-Approach
Yunyong Ko, Kyung-Jae Cho, and Sang-Wook Kim
Information Sciences (SCIE Journal, 2018)
- [2] Influence Maximization in Social Networks: A Target-Oriented Estimation
Yunyong Ko, Dong-Kyu Chae, and Sang-Wook Kim
Journal of Information Science (SCIE Journal, 2018)
- [1] Accurate Path-Based Influence Maximization in Social Networks
Yunyong Ko, Dong-Kyu Chae, and Sang-Wook Kim
WWW 2016 (*The ACM Web Conference*)
Short Paper (Acceptance Rate $\approx 21\%$)

Domestic Conference and Journal Papers

- [16] Performance Evaluation of Negative Sampling Methods in a Hyperedge Prediction Task
Daeun Lee, Song-Kyung You, Yunyong Ko, and Sang-Wook Kim
KIPS Spring Conference 2024
- [15] Performance Evaluation of Graph and Hypergraph Learning for Real-world Dynamic Networks
Songkyung Yu, Daeun Lee, Yunyong Ko, and Sang-Wook Kim
KIPS Spring Conference 2024
- [14] Performance Evaluation: Knowledge Embedding Methods for Political Stance Prediction
Seongeun Ryu, Yunyong Ko, and Sang-Wook Kim
KIPS Spring Conference 2023 (*Received the Best Paper Award*)
- [13] CoAID+: COVID-19 News Cascade Dataset for Social Context Based Fake News Detection
Soeun Han, Yoonsuk Kang, Yunyong Ko, Jiwon Ahn, Yusim Kim, Seongsu Oh, Heejin Park, and Sang-Wook Kim
KIPS Transactions on Software and Data Engineering (KCI Journal, 2022)
- [12] Precision Switching for Efficient Matrix Factorization in Recommender Systems
Jae-Seo Yu, Yunyong Ko, Hong-Kyun Bae, Seokwon Kang, Yongseung Yu, Yongjun Park, and Sang-Wook Kim
KIPS Spring Conference 2021
- [11] COVID-19 Cascade Dataset for Fake News Detection
Soeun Han, Yoonsuk Kang, Yunyong Ko, Jiwon Ahn, Yusim Kim, Seongsu Oh, Heejin Park, and Sang-Wook Kim
KIPS Spring Conference 2021 (*Received the Best Paper Award*)
- [10] Parameter Sharding for Synchronous and Asynchronous Distributed Training
Johyung Jung, Utae Lim, Junhwan Park, Kibong Choi, Yunyong Ko, and Sang-Wook Kim
KSC 2020 (*Korea Software Congress*)
- [9] Parameter Sharding approaches for DNN Models with a Very Large Layer

Kibong Choi, Yunyong Ko, and Sang-Wook Kim
KIPS Fall Conference 2020

- [8] Performance Evaluation: Parameter sharding for Distributed Deep Learning
 Kibong Choi, Yunyong Ko, Hyungseung Jei, Hongchan Noh, and Sang-Wook Kim
KCC 2019 (*Korea Computer Congress*)
- [7] Inter-Node Communications Methods for Distributed Deep Learning
 Kibong Choi, Yunyong Ko, and Sang-Wook Kim
KSC 2018 (*Korea Software Congress*)
- [6] A Diffusion Model for Influence Maximization in Selecting Advertisement Agent
 Suk-Jin Hong, Yunyong Ko, Sang-Wook Kim, and Gyehwan Park
CSMS 2018 (*KISM Workshop on Convergent & Smart Media Systems*)
- [5] Accurate Ad-Effect Estimation Method based on Relevance between User and Item
 Suk-Jin Hong, Yunyong Ko, Sang-Wook Kim, and Gyehwan Park
KOCON 2018 (*Korea Contents Association*)
- [4] Effective Ad-Effect Maximization Exploiting User's Support and Share
 Suk-Jin Hong, Yunyong Ko, Sang-Wook Kim, and Gyehwan Park
KIPS Spring Conference 2018
- [3] Accurate Trust Prediction Based on the Uninteresting User Concept
 Jonghyun Kim, Seungwon Yun, Yunyong Ko, Jangwan Koo, and Sang-Wook Kim
KSC 2017 (*Korea Software Congress*)
- [2] Fast Influence Maximization in Social Networks
Yunyong Ko, Kyung-Jae Cho, and Sang-Wook Kim
Journal of KIISE (KCI Journal, 2017) (*Received the Best Paper Award*)
- [1] Efficient CELF Algorithm for Community-based Influence Maximization in Social Networks
Yunyong Ko, Kyung-Jae Cho, and Sang-Wook Kim
KCC 2017 (*Korea Computer Congress*) (*Received the Best Presentation Award*)

INVITED TALKS

KHAN: Knowledge-Aware Hierarchical Attention Networks for Accurate Political Stance Prediction

- Invited Talk @ Electronic & Information Research Information Center (EIRIC), Sep. 2023

Not All Layers Are Equal: A Layer-Wise Approach Towards Large-Scale DNN Training

- Poster Session @ Hyundai Vision Conference (HVC), Aug. 2023
- Invited Talk @ METU-HYU Joint Workshop, Dec. 2022

Basic Concept of Distributed Deep Learning with PyTorch Tutorials

- Invited Talk @ Medical AI Korea, Oct. 2021

Efficient and Effective Influence Maximization in Social Networks: A Hybrid-Approach

- Invited Talk @ Waseda-UMS-Hanyang-UKM (WUHU) Joint Workshop, Dec. 2018
- Invited Talk @ NAVER Corp., Feb. 2018

PROFESSIONAL SERVICES

Track Co-Chair

- The ACM Symposium on Applied Computing (**ACM SAC**) 2023, 2024

Conference Reviewer

- The ACM Web Conference (**WWW**) 2023, 2024
- The ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**ACM KDD**) 2021, 2022, 2024
- The IEEE International Conference on Data Mining (**IEEE ICDM**) 2022, 2023
- The IEEE International Conference on Big Data (**IEEE BigData**), GTA3 Workshop 2023

- The AAAI International Conference on Artificial Intelligence (**AAAI**) 2021
- The ACM Symposium on Applied Computing (**ACM SAC**) 2022, 2023

Journal Reviewer

- The IEEE Transactions on Neural Networks and Learning Systems (**IEEE TNNLS**) 2023
- The Journal of Supercomputing 2023

PATENTS

Granted Patents

- Asymmetric Centralized Training for Distributed Deep Learning
Registration Number: **KR10-2555268** Jul. 2023
- Multi-State Diffusion Model Using Interest, Intimacy, and Share Tendency
Registration Number: **KR10-2332348** Nov. 2021
- Accurate Ad-Effect Estimation Method based on Relevance between User and Item
Registration Number: **KR10-2144122** Aug. 2020
- Hybrid Approach for Influence Maximization in Social Networks
Registration Number: **KR10-1810864** Dec. 2017

Filed Patents

- Personalized News Recommendation Technique using Category-guided Intent Disentanglement and Consistency-based News Representation
Application Number: **KR10-2023-0196006** Dec 2023
- Political Stance Prediction Method and System using Political Knowledge Graphs and Hierarchical Attention Networks
Application Number: **KR10-2023-0059346** May 2023
- Method and System for Adjusting the Learning Rate Differentiated by Layer for Large Scale Data Parallel Based Deep Learning
Application Number: **KR10-2022-0075800** June 2022

REFERENCES

- Hanghang Tong**, *Associate Professor* (Postdoc. Advisor) htong@illinois.edu
Department of Computer Science, University of Illinois at Urbana-Champaign (UIUC)
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College of Information Sciences and Technology, The Pennsylvania State University (PSU)
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